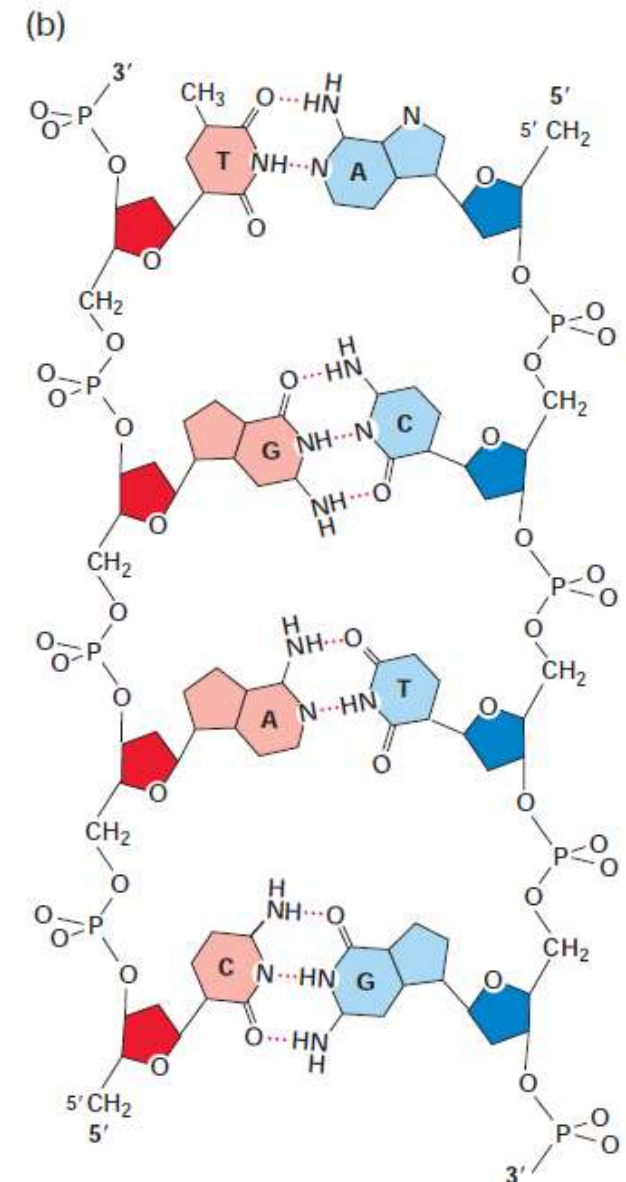
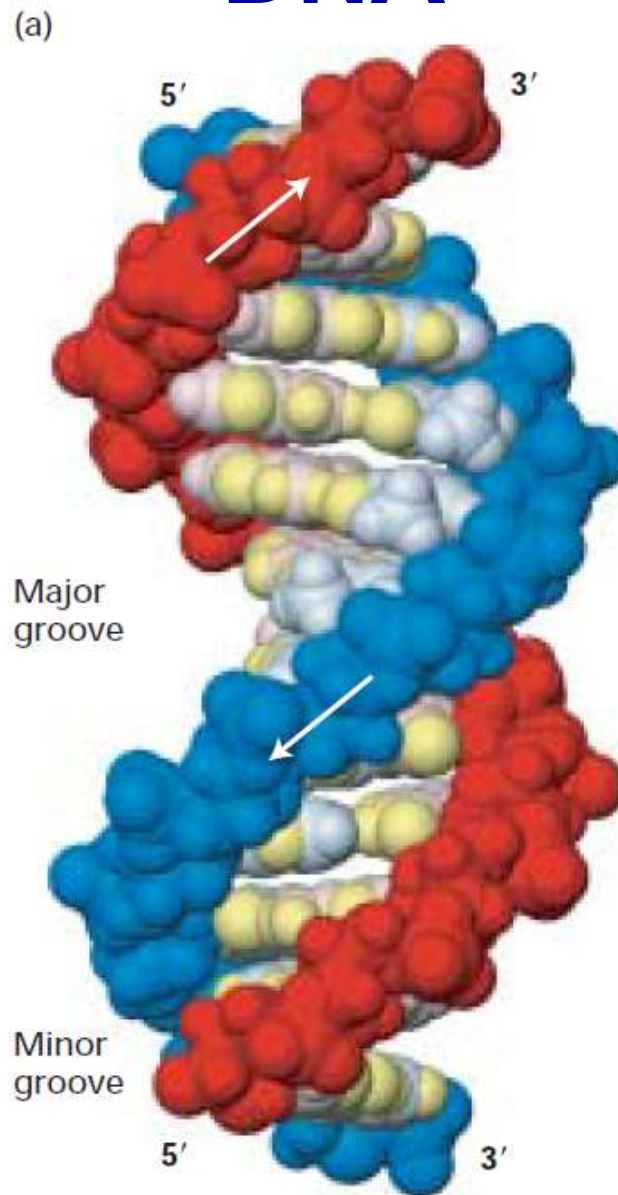


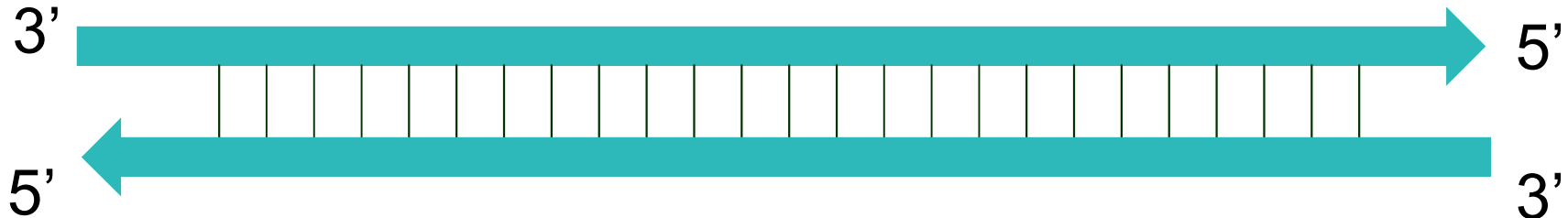
DNA

- DNA is a Molecule made by two polymeric chains, waived in the famous double-helix structure.
- Each chain is made by a combination of 4 monomers, the nucleotides.
- Each monomer is spaced by .34 nanometers
- Each turn of the helix contains about 10 bases
- A full DNA chain could be several cm long

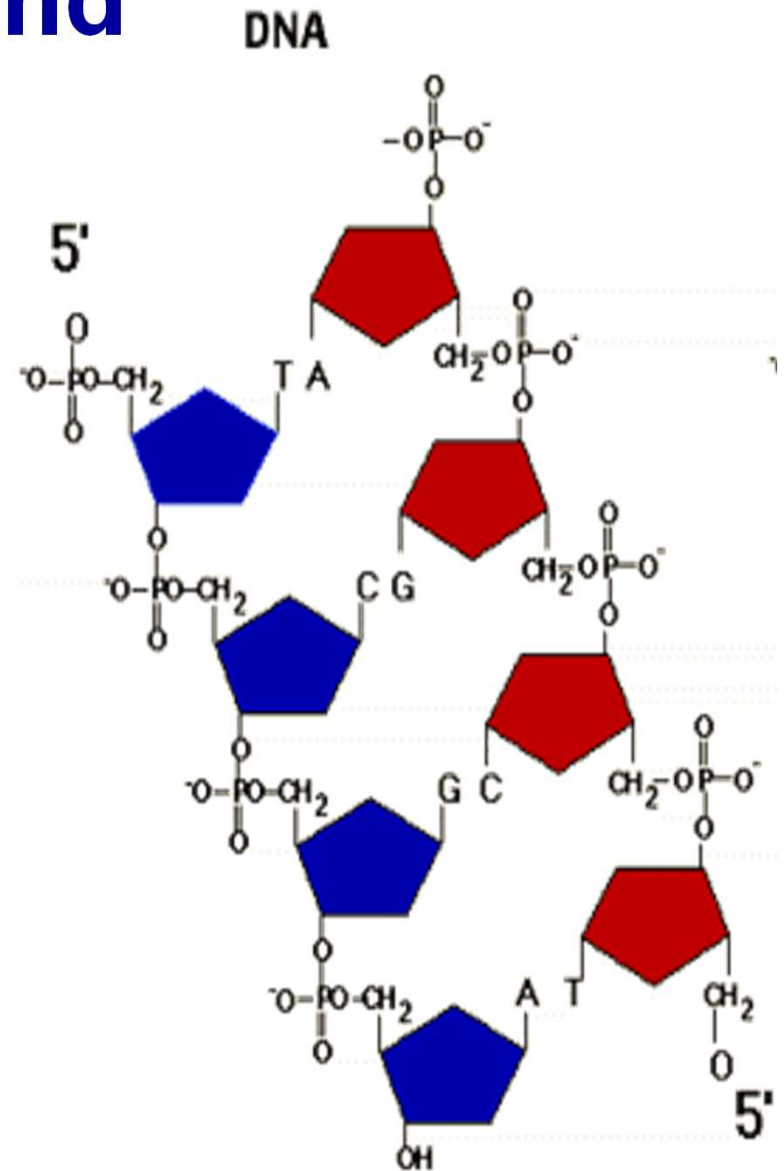
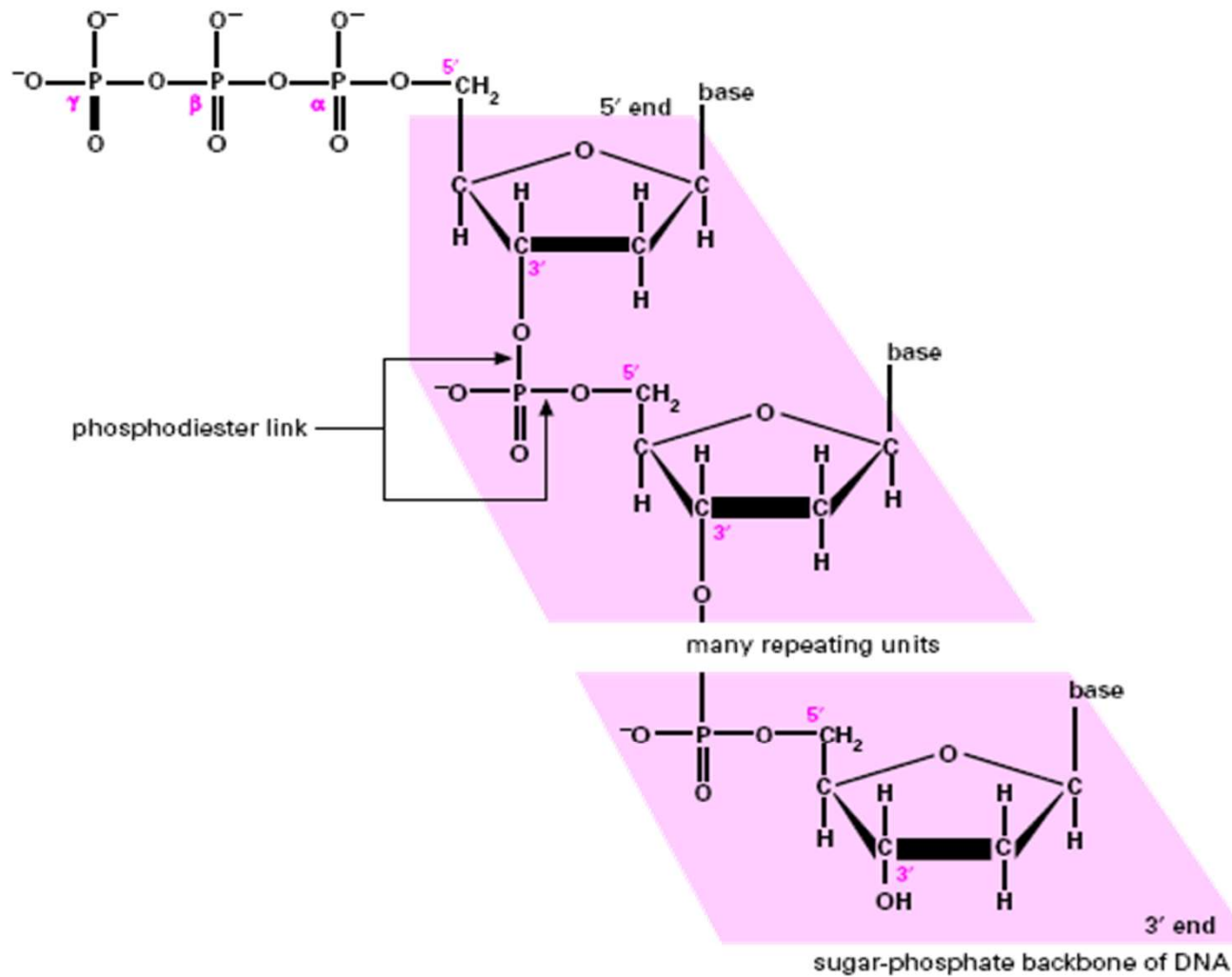


Orientation of the DNA

- A “single strand DNA” is not a symmetric molecule, it has a direction (i.e. 3'→ 5')
- The DNA double helix is made by two paired single-strand molecules proceeding in opposite directions

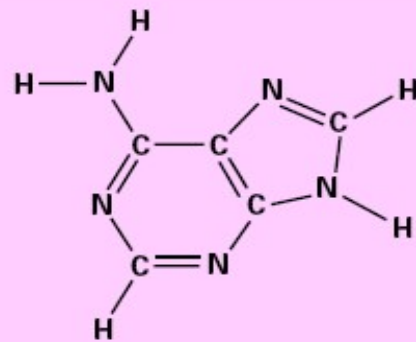


Phosphate bond

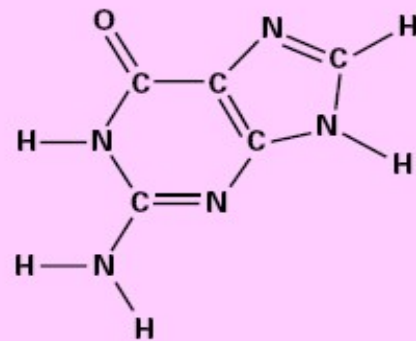


Bases

purines

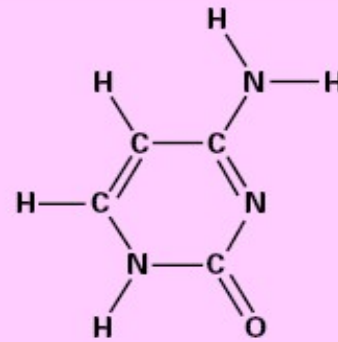


adenine (A)

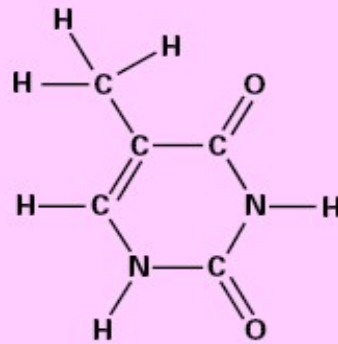


guanine (G)

pyrimidines



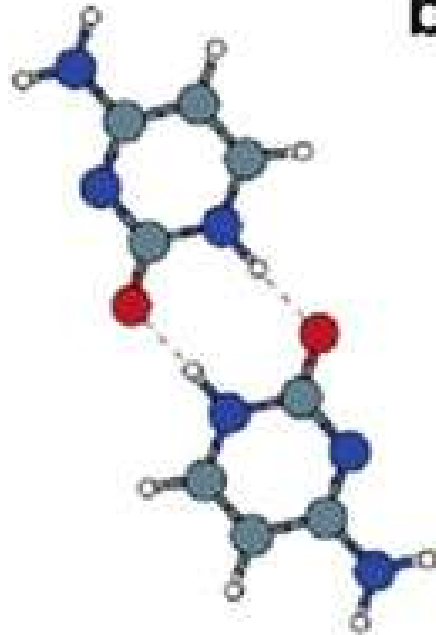
cytosine (C)



thymine (T)

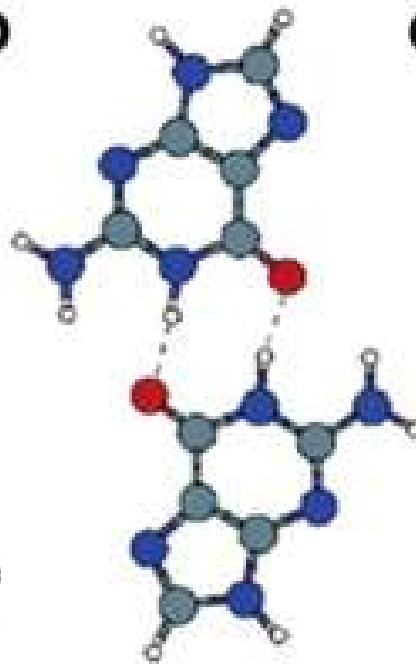
Hydrogen Bonds

a



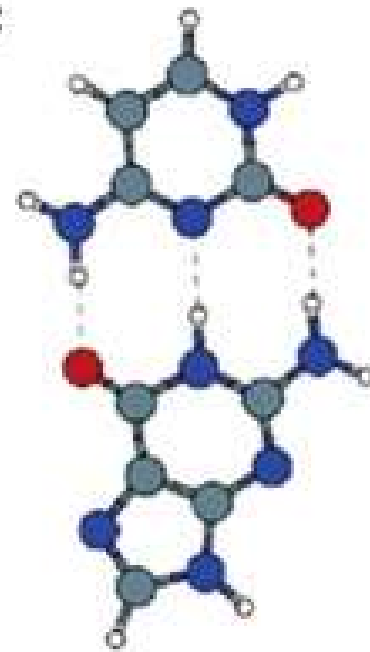
C-C

b



G-G

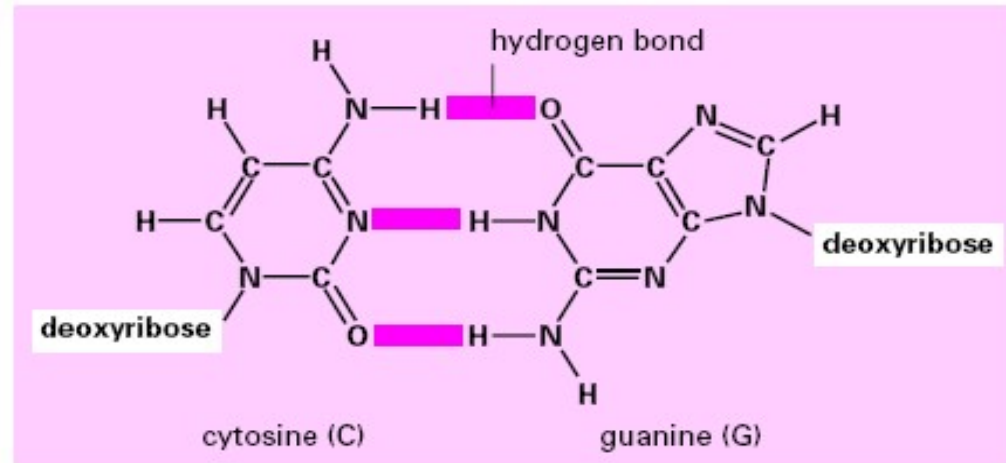
c



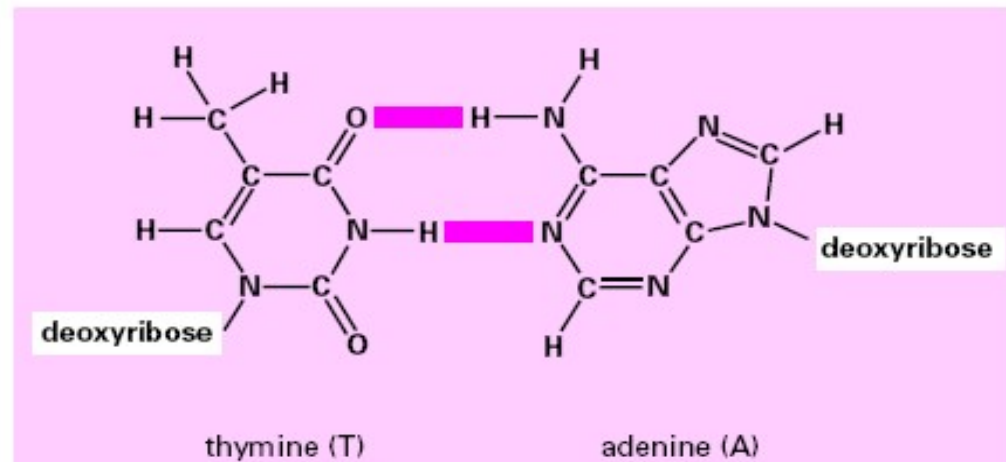
C-G

Hydrogen bonds

cytosine-guanine
base pair (C≡G)



thymine-adenine
base pair (T=A)



DNA Complementarity

- $A \rightarrow T$, $G \rightarrow C$



- So, the complementary strand of ATTAGGCGC is GCGCCTAAT (negate and reverse)

Ambiscript notation of DNA

	Pyrimidine (Light)	Purine (Heavy)
Strong	 Cytosine	 Guanine
Weak	 Thymine	 Adenine

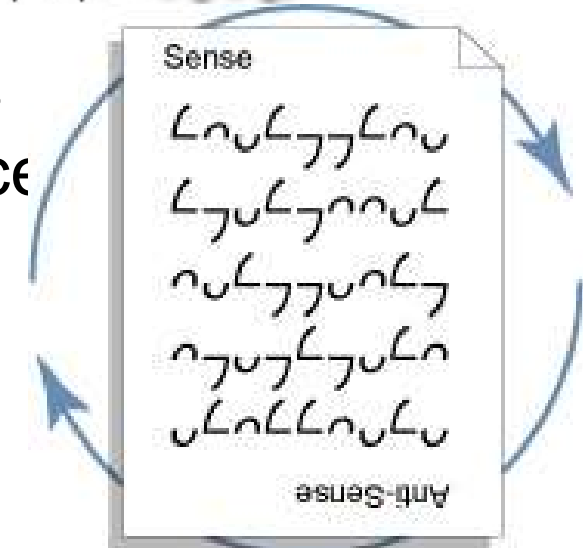
- Easier to read

A GATTTACTAT
CTATGTGACT
GCTGGATGCG
TATGCATGAT

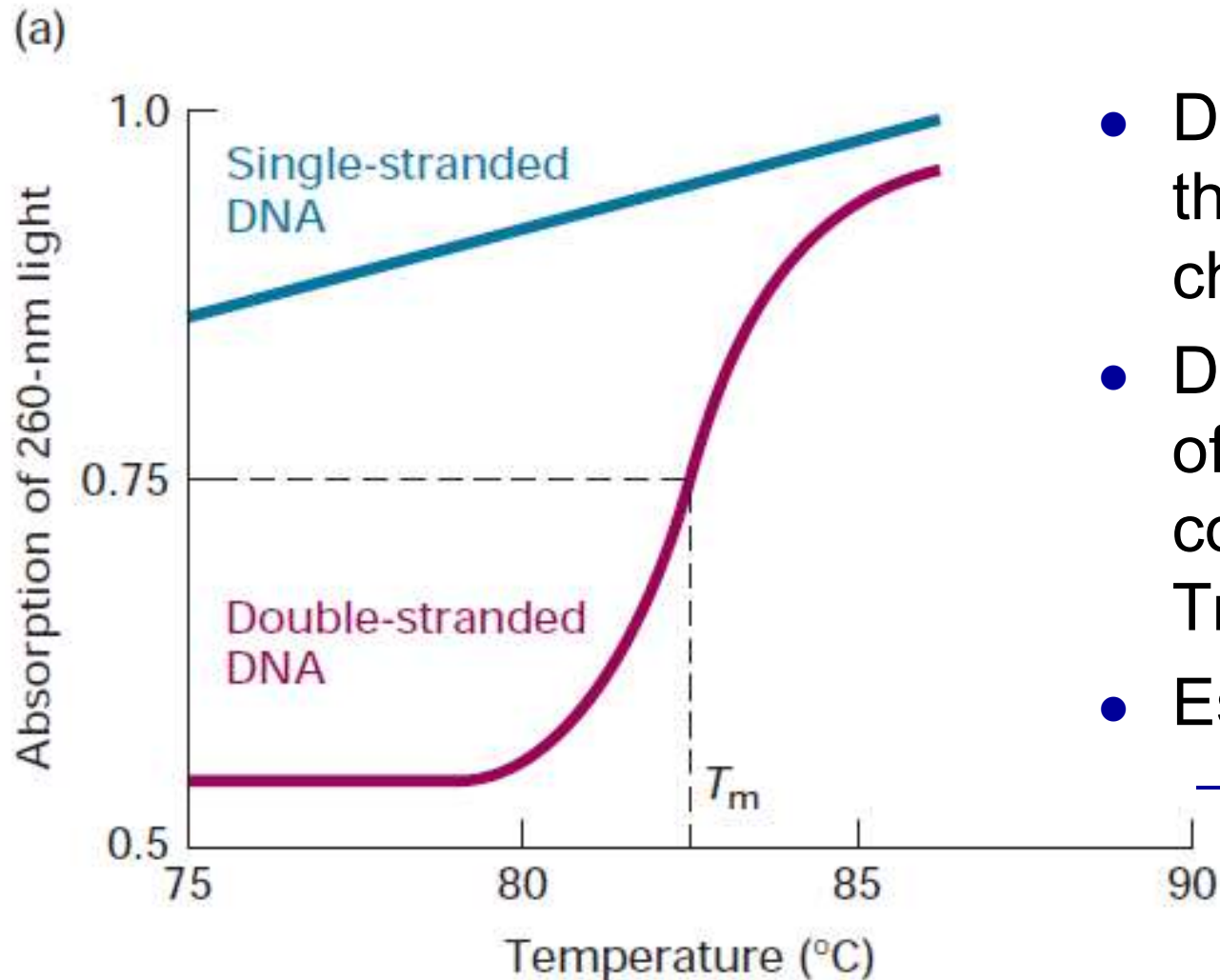
B ԴԱԴԴԴԴԴԴ
ԼԴԴԴԴԴԴԼ
ԴԼԴԴԴԴԴԼԼ
ԴԴԴԼԴԴԴԴԼ

C 

- Easier to convert to its complimentary sequence (just turn the page)



Melting temperature



- Depends on the length of the chains (longer chains=higher T_m)
- Depends on the percentage of AT couples respect to GC couples. (more CGs =higher T_m)
- Es. Calculator:
 - http://www.biophp.org/minitools/melting_temperature/demo.php

Thermodynamic properties of DNA

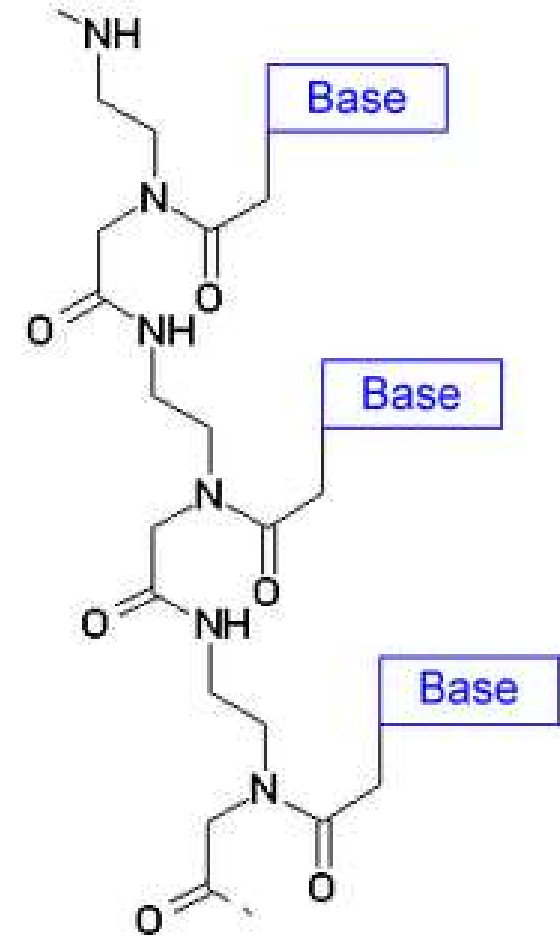
- At room temperature, DNA is made by two complementary chain in the stable double-helix configuration
- At higher temperatures (95 °C) the two chains separate (Denaturation)
- When the temperature decreases, two complementary chains find each other and sticks, returning to the double helix configuration (Hybridization)



Scaling up 1 million times, It is like a needle (2cm) in a lake (10km) that finds his pair by itself

Not only DNA and RNA

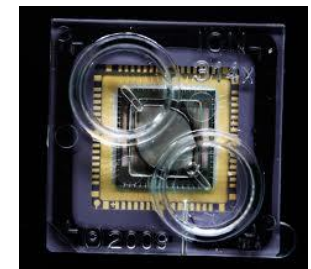
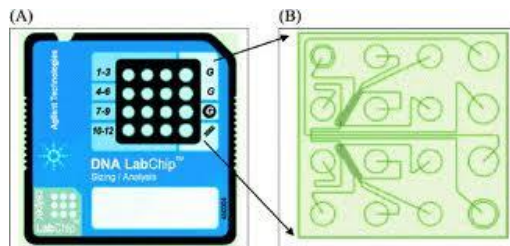
- PNA (peptide nucleic acid)
- An artificial “nucleic acid”, has the same 4 bases as DNA and RNA, but uses a peptide (proteic) chain as backbone
- Hybridizes with DNA and RNA
- T_m much higher than DNA-DNA bond (more stable) due to the neutral backbone



Biochips to test DNA

1

Biology	Task	Information technology
Capillary electrophoresis	Measure the size of a sequence	Length(x)
DNA Microarray	Find the occurrence of a sequence	Find(x) in file
PCR	Replicate a sequence	Copy + Paste
DNA Sequencing	Acquire a sequence	File read





DNA Microarray:

Find the occurrence of a sequence in a file

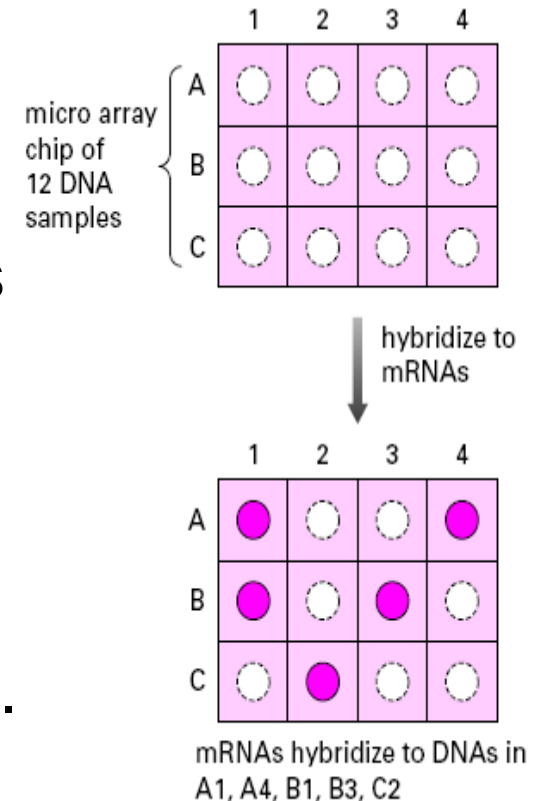
Find(x) in file

2

- Gene expression
 - X is any sequence in the human genome, File is RNA taken from a cell.
 - If X is found, the cell is executing that portion of genetic code.
- Comparative research on gene functions
 - 2 populations differing from 1 characteristic (i.e. eye color) == File1 and File2
 - Any X found on File1 and not on File2 may be related to the functions
- Diagnose an infectious disease:
 - X is a sequence typical of a pathogens, File is the sample taken from a patient
 - if X is found, the pathogen is present
- Genetic disease or predispositions
 - X is the mutated sequence causing the disease
 - If X is found, the patient is at risk

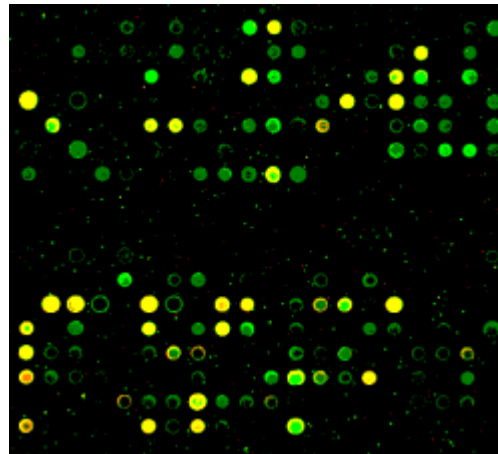
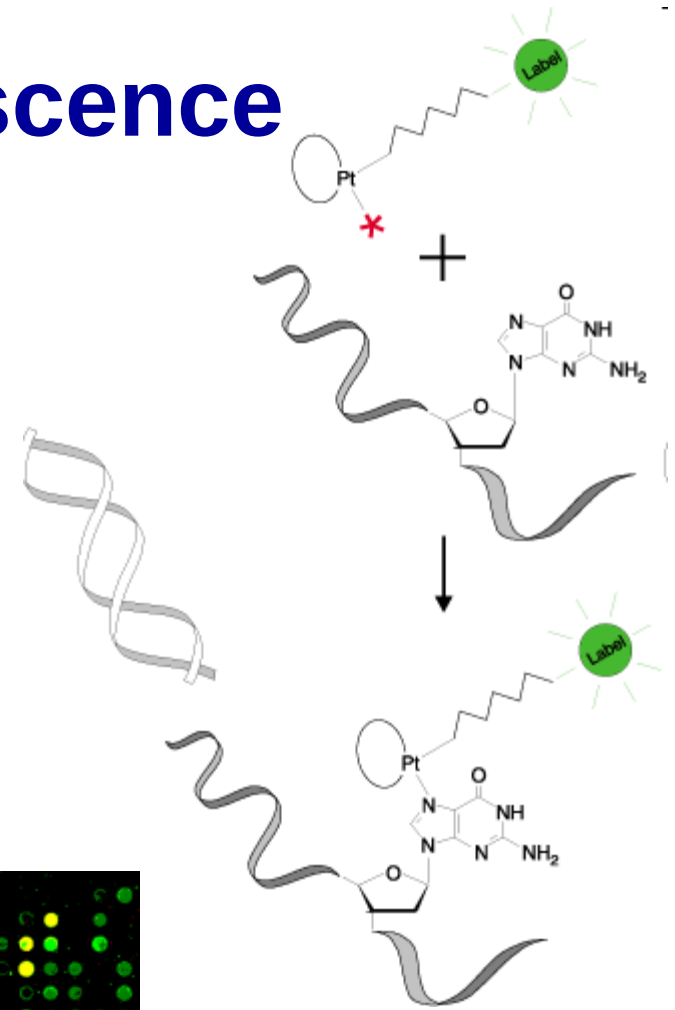
Principle of the microarray

- Known DNA is attached (i.e. By spotting) on the chip surface, at known positions
- An unknown sample containing DNA is placed on the chip.
- Wait for the DNA to hybridize to its complementary and then wash.
- Acquire the map of positions where DNA sticks (i.e. optically or electrically).
- Result: in the sample there where the sequences correspondent to the active spots



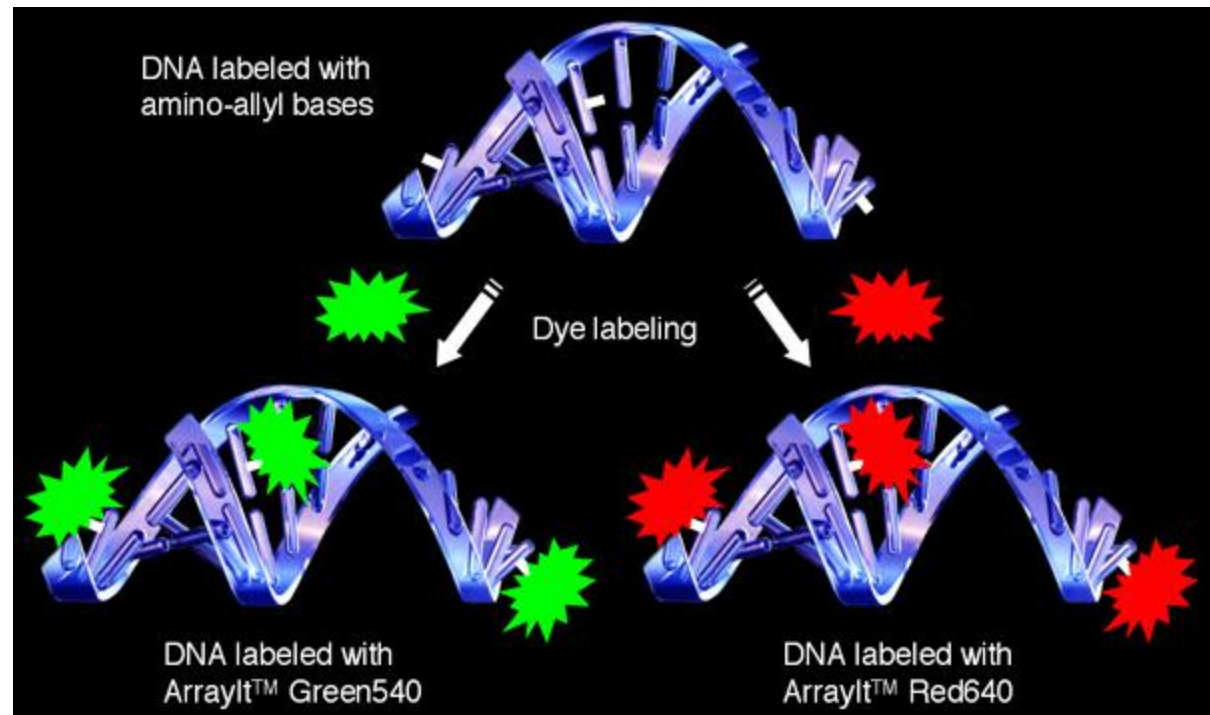
Detection by fluorescence

- Before the experiment all the DNA in the sample is made fluorescent by chemically attaching a fluorescent molecule to one of its ends
- After washing, the chip is observed under a fluorescence microscope. Only where the DNA hybridized a light signal is detected



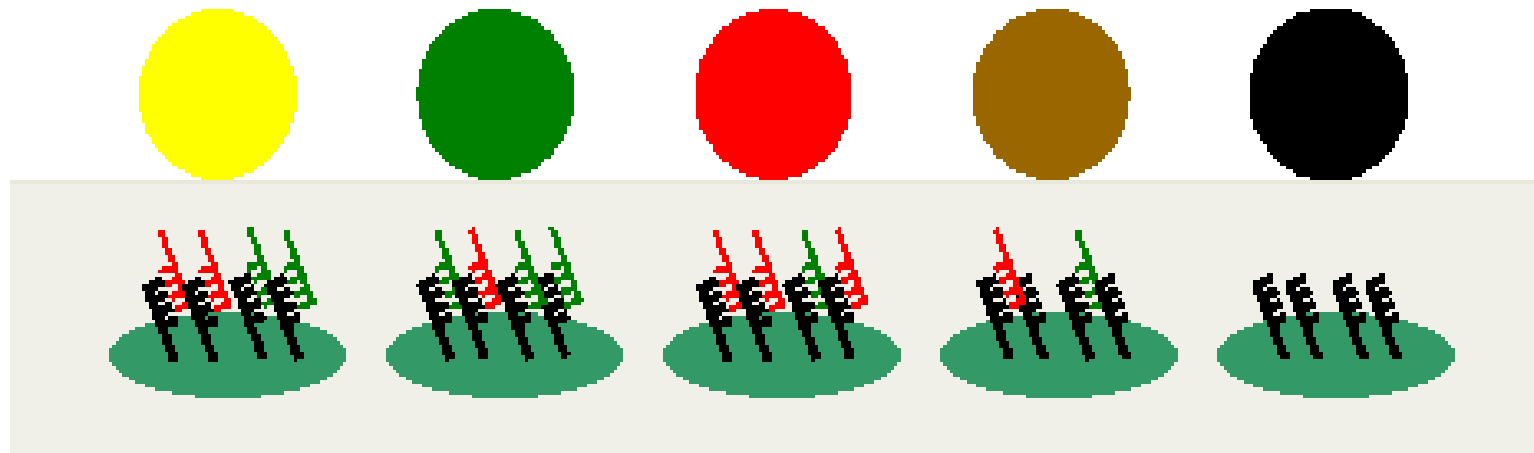
Revelation with 2 (or more) colors

- Different samples are attached to different fluorophores (with different emission wavelength)



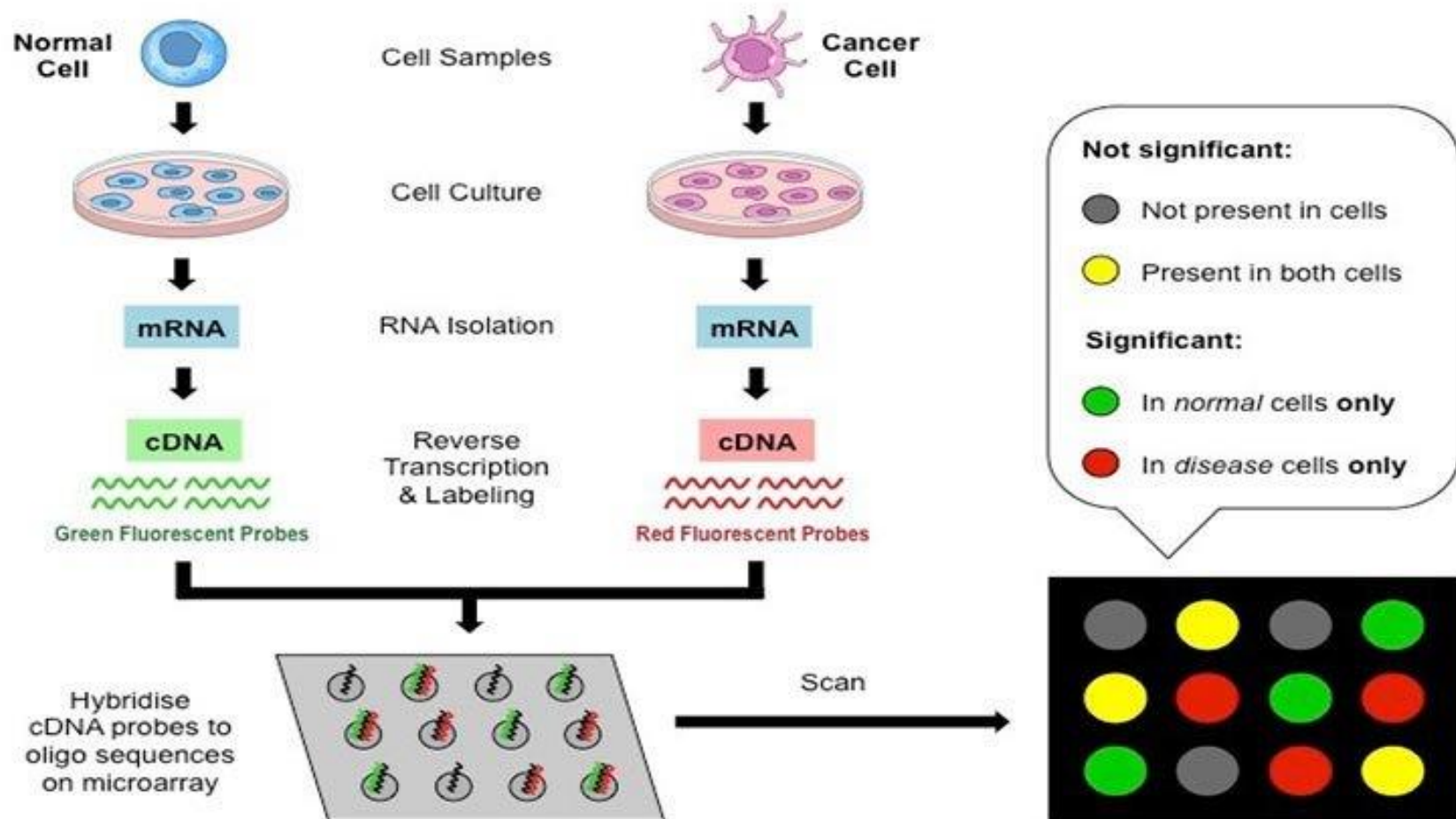
Two colors: Comparative tests

- One sample is labeled with green fluorophore, the other is labeled with red.
- Results:
 - Green: DNA found in sample 1
 - Red: DNA found on sample 2
 - Yellow: DNA found on both samples
 - Black: DNA not present in any sample

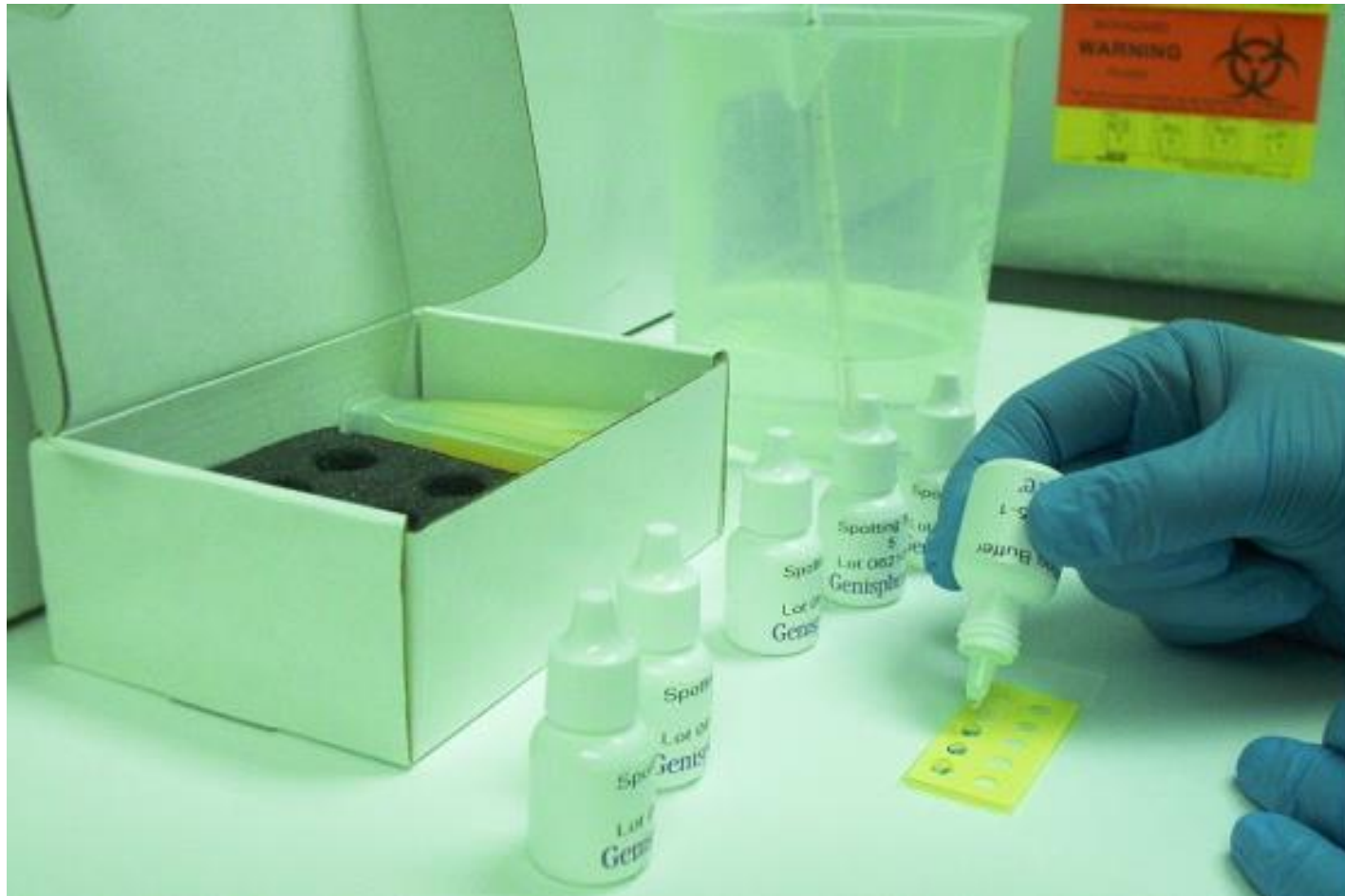


Reverse Transcriptase

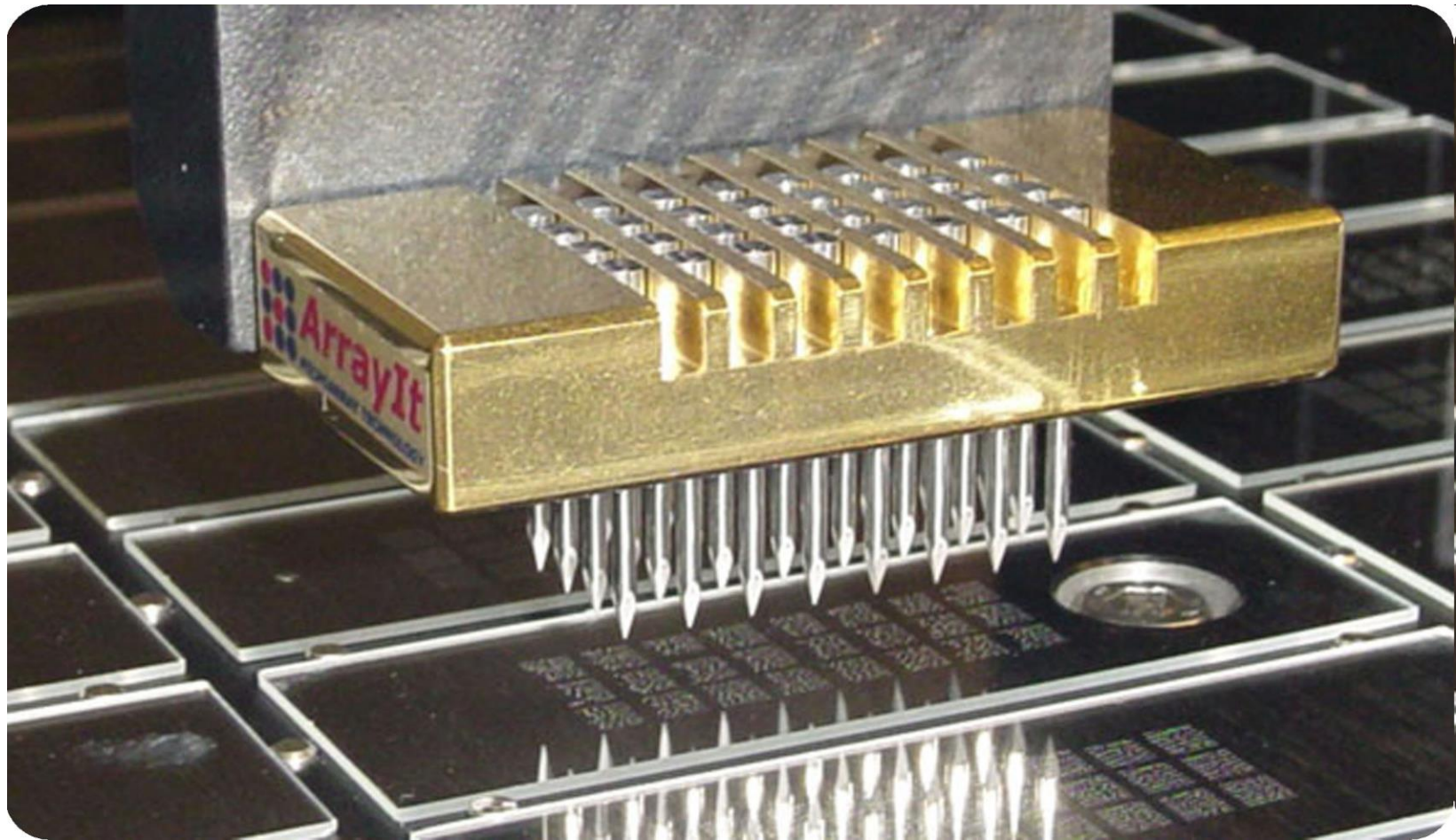
- The Central dogma of molecular biology states that: “DNA makes RNA, and RNA makes protein”
- Not completely true: some viruses (retroviruses) are capable to make DNA copies from RNA, maintaining the same sequence of bases
- The enzyme responsible for this operation is the *reverse transcriptase*
- To apply to RNA the same techniques used for DNA, initially RNA is converted to its DNA complement and then tested as DNA



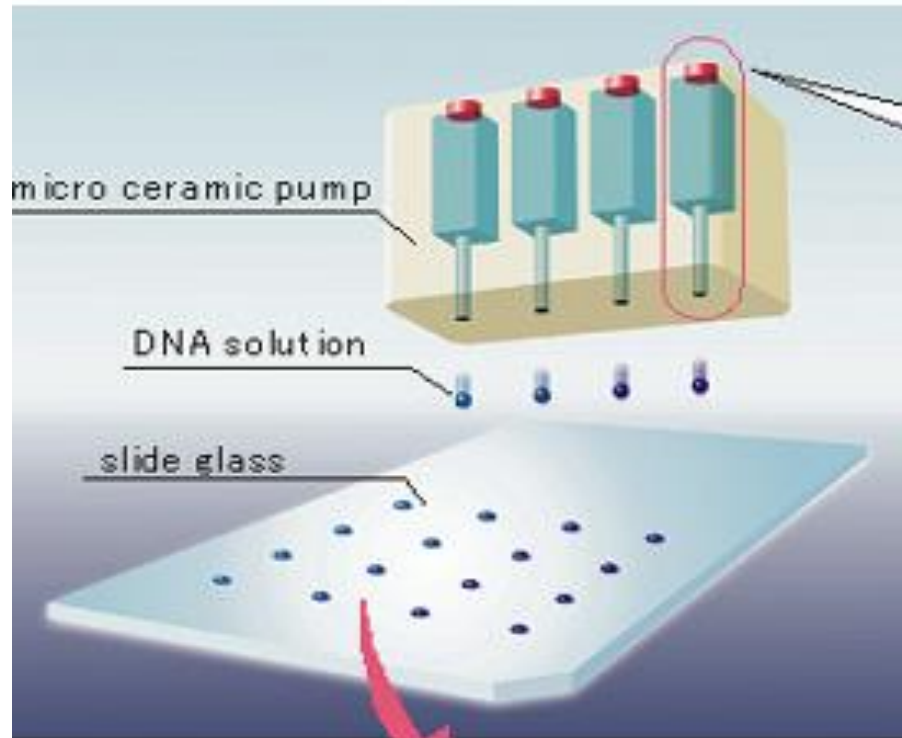
Manual Spotting



Automatic spotter

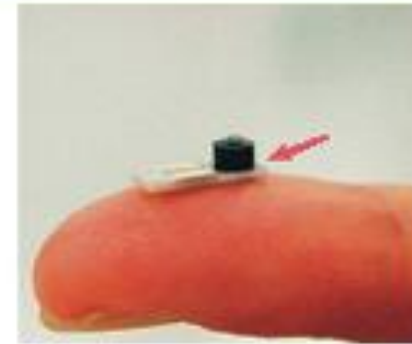


Automatic spotter: Principle

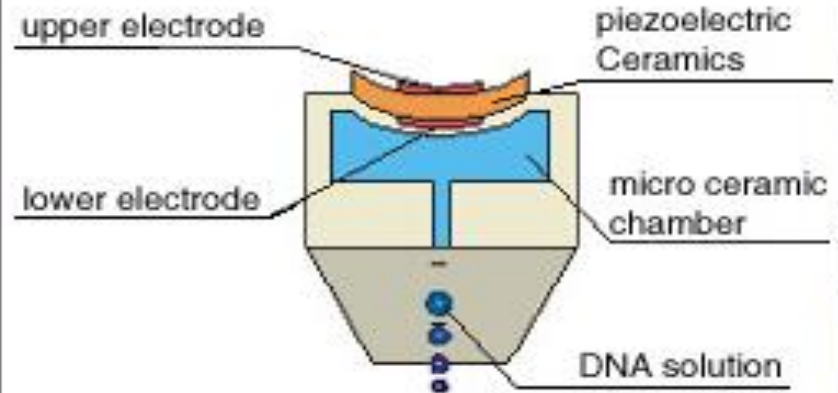


DNA
microarray

MICRO CERAMIC PUMP



● micro ceramic pump

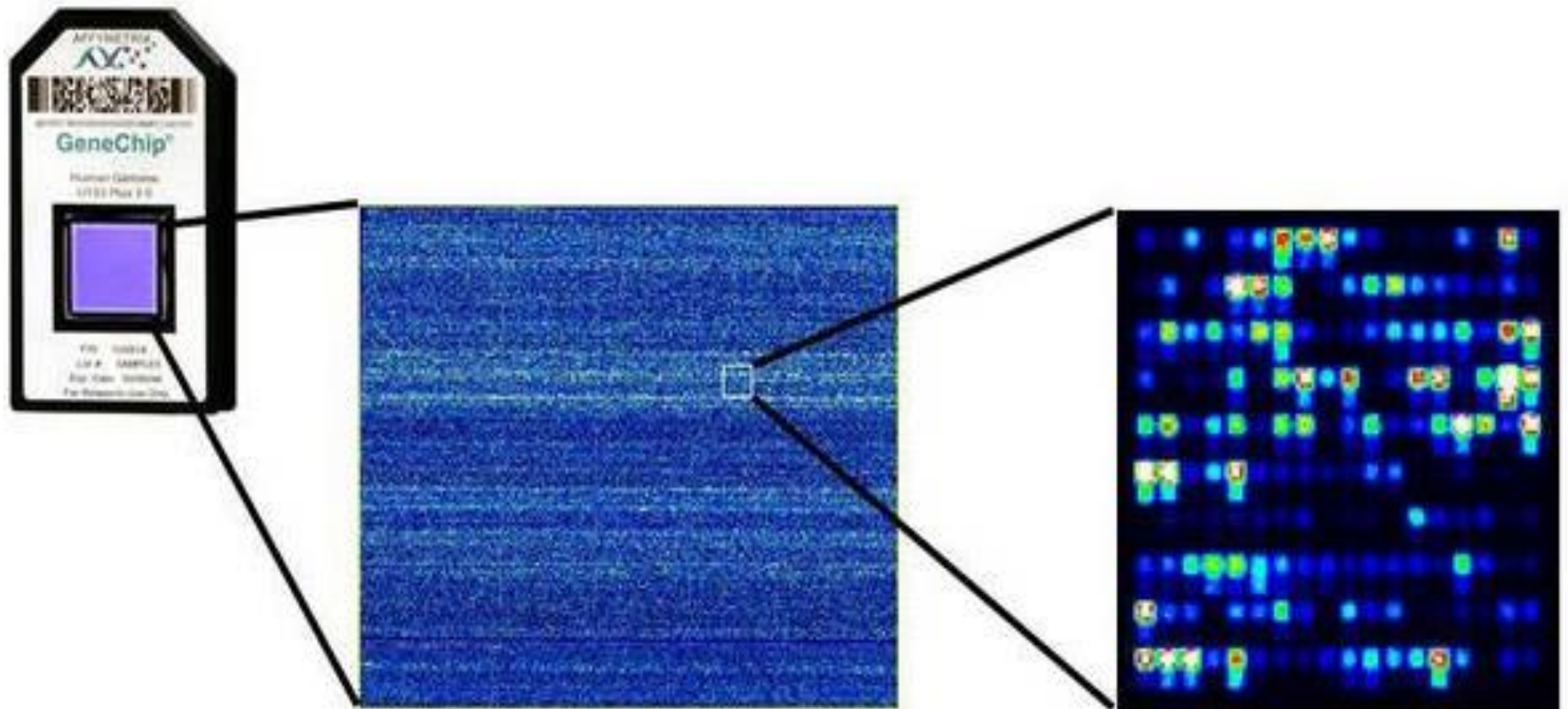


High density microarrays

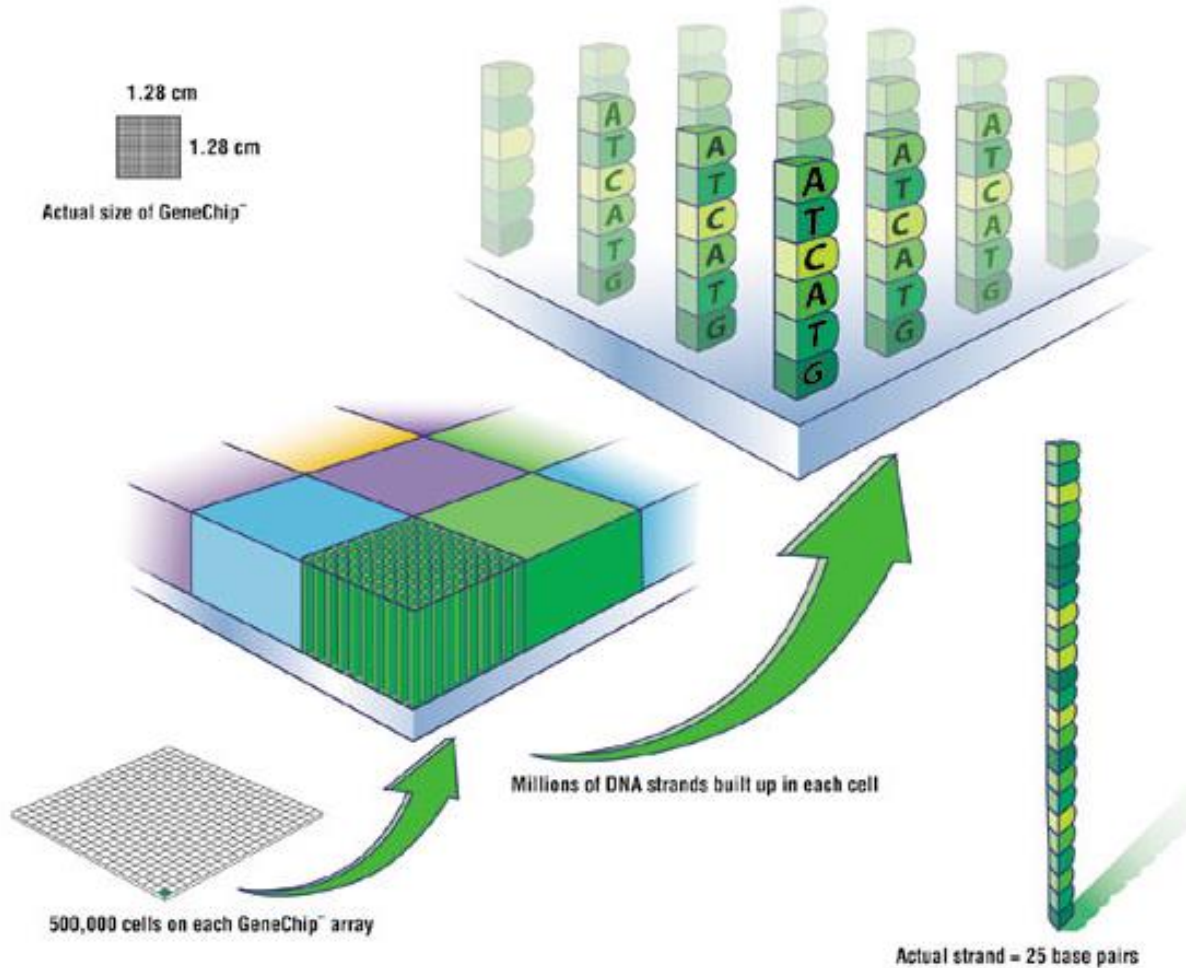
- Microarrays containing thousands (on millions) of spot.
- It is possible to have a full genome and all its variations (SNPs) on a single chip



Affymetrix chip



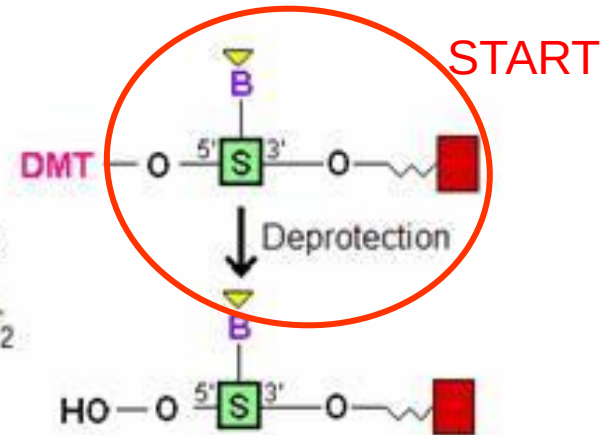
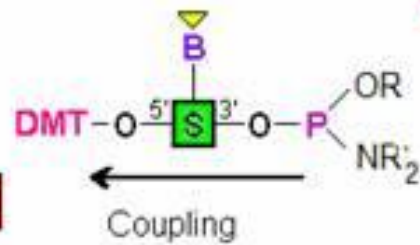
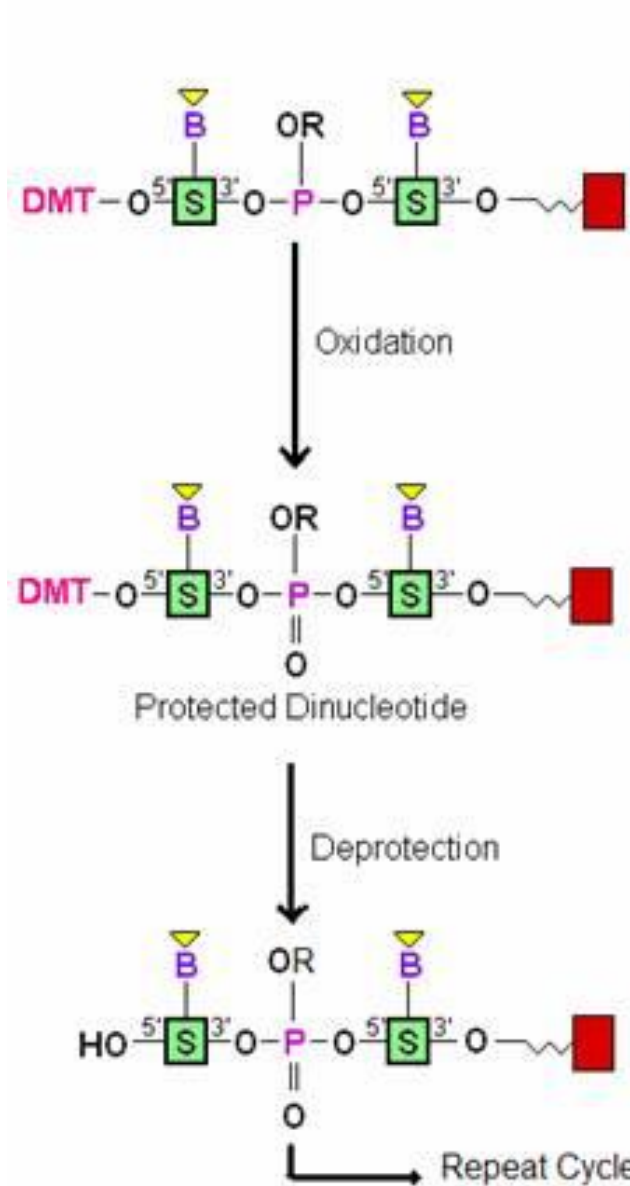
Detail of microarray



Artificial DNA synthesis

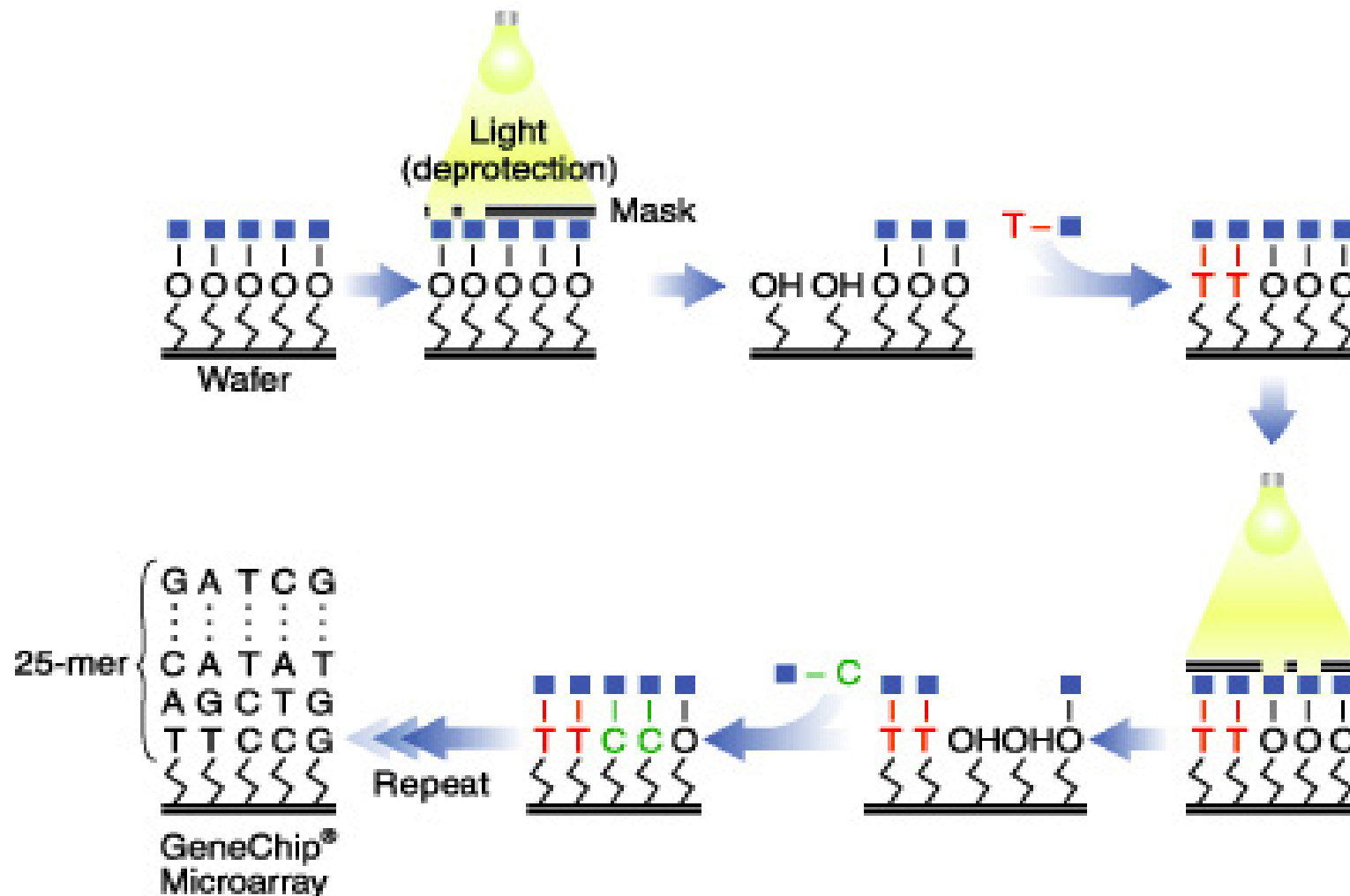
- We start with a base linked on its 3' end on a solid medium (i.e. Porous column, Bead, chip). The 5' end is protected by a 'capping' (DMT=Dimethoxytrityl) , preventing the formation of other bonds.
- 1: DMT is chemically removed (i.e. with trichloroacetic acid TCA), the 5' carbon becomes reactive. The chemical agent is then washed away
- 2: A solution containing pure basis (only A or T or C or G) is put on the substrate. Those new bases are already 'capped' in 5'. These bases react in 3' with the 5' of the DNA chain
- 3: The bond is fixed by oxidation of the phosphate
- The cycle is repeated from step1, reactivating the new end by de-capping
- When the desired sequence is finished, the DNA is 'harvested' cutting the original 3' bond to the medium.
- The final DNA sequence depend on the order of the bases as chosen in step 2

Solid Phase Nucleic Acid Synthesis



-  Solid CPG support
-  Ribose/deoxyribose sugar
-  Protected base

Photolithographic process



Example:

- Dots: D1= A-A-T... D2=T-A-G... D3=T-G-T... D4...
- Steps:

		Nucleotide flown	Mask	D1	D2	D3
1 st round	{	A	_ _ _	oA	o	o
		C	_ _ _ _	oA	o	o
		G	_ _ _ _	oA	o	o
		T	_ _ _ _	oA	oT	oT
2 nd round	{	A	_ _ _ _	oAA	oTA	oT
		C	_ _ _ _	oAA	oTA	oT
		G	_ _ _ _	oAA	oTA	oTG
		T	_ _ _ _	oAA	oTA	oTG
3 rd round	{	A	_ _ _ _	oAA	oTA	oTG
		C	_ _ _ _	oAA	oTA	oTG
		G	_ _ _ _	oAA	oTAG	oTG
		T	_ _ _ _	oAAT	oTAG	oTGT

Illumina bead-array

- Instead of growing DNA on the array, it is synthesized in lots with high quality and purity.
- DNA is attached on the surfaces of small balls (Beads) 3 micron diameter. This step is done in lots for high quantities..
- Different beads are mixed to create a pool
- Beads are randomly placed on the chip
- Each chip is different from others. It has a specific map, provided with the interpretation SW



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